

أسئلة سنوات

فاينل

لجنة سنافر البوليتكنك - بسواعدنا نبنيها

اسم المادة

الكيمياء العامة



تواصل معنا

www.snafermuslimengineer.weebly.com

[f /groups/Smurfs.On.The.Way](https://www.facebook.com/groups/Smurfs.On.The.Way)

[f BESOWA3EDNA.NABNEHA](https://www.facebook.com/BESOWA3EDNA.NABNEHA)

[f Muslimengineerpage](https://www.facebook.com/Muslimengineerpage)

0.000 C/mole e⁻

Transfer your answer to the table below in CAPITAL LETTERS

Question	1	2	3	4	5	6	7	8	9	10	11	12	13
Answer					D	E	C						

Question	14	15	16	17	18	19	20	21	22	23	24	25
Answer									E		E	E

Choose the most correct answer for each of the following questions:

1. Consider the unbalanced redox equation:

$\text{CH}_3\text{OH}_{(l)} + \text{Cr}_2\text{O}_7^{2-}(\text{aq}) + \text{H}^+(\text{aq}) \rightarrow \text{CH}_2\text{O}_{(\text{aq})} + \text{Cr}^{3+}(\text{aq}) + \text{H}_2\text{O}_{(l)}$. Which of the following sets of numbers will balance the equation?

- A) 3, 1, 8, 3, 2, 7 B) 1, 3, 8, 3, 2, 4 C) 8, 1, 3, 4, 3, 2 D) 3, 8, 1, 2, 3, 7
E) 3, 1, 8, 3, 2, 4

2. In which of the following does sulfur have an oxidation number of +7?

- A) H_2SO_3 B) $\text{H}_2\text{S}_2\text{O}_3$ C) H_2SO_4 D) $\text{H}_2\text{S}_2\text{O}_8$ E) S_2F_{10}

Examine the data on the right. The strongest oxidizing agent among these substances is:

- A) Sr^{2+} B) Cr^{2+} C) Co^{2+}
D) Zn^{2+} E) Fe^{2+}

half-reduction reactions	E° (V)
$\text{Sr}^{2+} + 2\text{e}^- \rightarrow \text{Sr}$	-2.89
$\text{Cr}^{2+} + 2\text{e}^- \rightarrow \text{Cr}$	-0.913
$\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe}$	-0.447
$\text{Co}^{2+} + 2\text{e}^- \rightarrow \text{Co}$	-0.280
$\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn}$	-0.760

The mass (in g) of Cu (63.55 g/mol) that can be deposited by the passage of 12.0 A for 25.0 min through $\text{CuSO}_4(\text{aq})$ solution equals:

- A) 5.93 B) 16.59 C) 6.91 D) 19.36 E) 14.38

A flask containing 0.124 mol of A and was allowed to react according to the equation $\text{A}_{(\text{g})} \rightarrow \text{B}_{(\text{g})}$, the data

Time (s)	0.00	10.0	20.0	30.0	40.0
Moles of A	0.124	0.110	0.088	0.073	0.060

on the right were obtained for moles of A as the reaction proceeded. The average rate of disappearance of A between 10 and 20 seconds equals:

- A) 1.9×10^{-3} B) 2.2×10^{-3} C) 3.1×10^{-3} D) 1.75×10^{-3}
E) 1.50×10^{-3}

In $[N_2O_5]$ and the time, and $\Delta \ln[N_2O_5]$ was -1.16 during a time period of 2030 s. The value of the rate constant k equals:

- A) 7.25×10^{-4} B) 8.29×10^{-4} C) 7.73×10^{-4} D) 6.44×10^{-4} E) 5.71×10^{-4}

7. During the experimental determination of the rate law for the reaction $A + B \rightarrow P$, it was found that the rate increased by a factor of 9 when the concentration of B was tripled, this reaction is -----order in B

- A) Zero B) first C) second D) third E) forth

8. Determine the rate law expression for the reaction $2A + B_2 + C \rightarrow A_2B + BC$, using the kinetic data provided on the table:

Run	$[A]_0$	$[B_2]_0$	$[C]_0$	Initial rate (M/s)
1	1.2	1.0×10^{-2}	1.0	1.6×10^6
2	2.4	1.0×10^{-2}	1.0	1.6×10^6
3	1.2	2.0×10^{-2}	1.0	3.2×10^6
4	1.2	1.0×10^{-2}	2.0	6.4×10^6

- A) $R = k[A][C]^2$ B) $R = k[A][B_2]^2$
 C) $R = k[B_2]^2[C]^2$ D) $R = [A]^2[B_2]^2$
 E) $R = k[B_2][C]^2$

The rate constant k for the reaction $CH_3N \equiv C \rightarrow CH_3C \equiv N$ equals $6.29 \times 10^{-4} \text{ s}^{-1}$, if the $[CH_3N \equiv C]_0$ is $1.00 \times 10^{-3} \text{ M}$, then its concentration after $3.5 \times 10^3 \text{ s}$ equals:

- A) 3.89×10^{-4} B) 1.11×10^{-4} C) 2.84×10^{-4} D) 2.08×10^{-4} E) 5.33×10^{-4}

9. For the second order reaction $A \rightarrow \text{products}$, 25 % of A were found to be consumed in 60.0. What is the second half life (in hr) when the initial concentration of A equals 1.0 M ?

- A) 466.2 B) 371.7 C) 600.6 D) 800.0 E) 1136.4

10. The rate constant for the reaction $A \rightarrow B$ is $0.039 \text{ M}^{-1}\text{s}^{-1}$. The final concentration of A after 45 s was 0.30 M , its initial concentration equals:

- A) 0.410 B) 0.463 C) 0.392 D) 0.480 E) 0.633

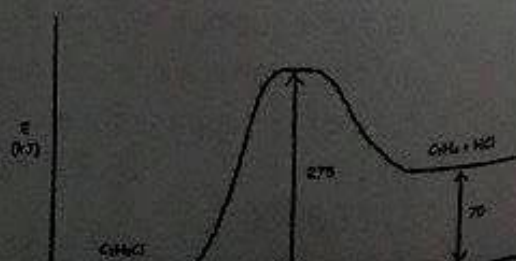
11. The rate constant k for a certain reaction was $3.7 \times 10^{-3} \text{ s}^{-1}$ at 25°C . If the activation energy for this reaction was 43.6 kJ/mol , then the value of rate constant for this reaction at 75°C equals:

- A) 5.7×10^{-2} B) 4.6×10^{-2} C) 3.7×10^{-2} D) 2.4×10^{-2} E) 4.9×10^{-3}

12. The rate equation for the reaction: $A \rightarrow B$ is $R = k[A]^0$. If the initial concentration of the reactant $[A]$ is 1.0 mol L^{-1} , the half-life period of the reaction is:

- A) $k/[A]$ B) $[A]/k$ C) $2[A]/k$ D) $[A]/2k$ E) $2k/[A]$

13. Carefully inspect the profile on the right for the reaction $C_2H_5Cl \rightarrow C_2H_4 + HCl$; what is the activation energy of the REVERSE reaction, and is the FORWARD reaction exothermic or endothermic?



- B) 205 kJ and exothermic reverse reaction
 C) 200 kJ and endothermic forward reaction
 D) 200 kJ and exothermic reverse reaction
 E) 205 kJ and endothermic forward reaction

15. Which one of the following statement is **INCORRECT** concerning a catalyst?

- A) A homogeneous catalyst is in the same phase as compared to the reactants.
 B) A heterogeneous catalyst is in a different phase as compared to the reactants.
 C) A catalyst enhances the rate by decreasing the E_a of the reaction.
 D) A catalyst is consumed during the reaction.
 E) A catalyst is chemically unchanged.

16. For the reaction $\text{CO}_{2(s)} \rightleftharpoons \text{CO}_{2(g)}$, if the standard enthalpy of sublimation equals 28.2 kJ/mol, then the standard entropy of sublimation (in J/mol) equals:

- A) 87.9 B) 233.0 C) 94.6 D) 209.5 E) 81.2

17. Use the data provided on the right to find ΔH for the combustion of 14.4 g of phenol ($\text{C}_6\text{H}_5\text{OH}$, 94.11 g/mol)

	$\text{C}_6\text{H}_5\text{OH}_{(l)}$	$+7\text{O}_{2(g)} \rightarrow$	$6\text{CO}_{2(g)} +$	$3\text{H}_2\text{O}_{(l)}$
ΔH°_f (kJ/mol)	-165.0		-393.5	-285.8

- A) -467.2 B) -824.0 C) -438.0 D) -561.3 E) -496.4

18. Use the data provided in the table below to find ΔH (kJ/mol) for the reaction: $\text{N} \equiv \text{N} + 3 \text{H}-\text{H} \rightarrow 2\text{NH}_3$

- A) -79 B) 75 C) -109 D) 79 E) -75



Bond	BE (kJ/mol)
$\text{N} \equiv \text{N}$	+945
$\text{H}-\text{H}$	+436
$\text{N}-\text{H}$	+388

19. Which one of the following reactions has a **POSITIVE** entropy change?

- A) $\text{BF}_3(g) + \text{NH}_3(g) \rightarrow \text{F}_3\text{BNH}_3(g)$
 B) $2\text{SO}_2(g) + \text{O}_2(g) \rightarrow 2\text{SO}_3(g)$
 C) $\text{H}_2\text{O}(g) \rightarrow \text{H}_2\text{O}(l)$
 D) $2\text{NH}_4\text{NO}_3(s) \rightarrow 2\text{N}_2(g) + 4\text{H}_2\text{O}(g) + \text{O}_2(g)$
 E) $\text{N}_2(g) + 3\text{H}_2(g) \rightarrow 2\text{NH}_3(g)$

For the reaction $2\text{CO}(g) + 2\text{H}_2(g) \rightarrow \text{CO}_2(g) + \text{CH}_4(g)$, if $\Delta H^\circ = -247.3$ kJ/mol and $\Delta S^\circ = -256.5$ J/mol K, the value of ΔG° for the reaction:

- A) -170.9 B) 175.6 C) 160.9 D) 181.6 E) 170.9

A solution was prepared by dissolving 50.0 mL of CHCl_3 in enough amount of acetone to produce 100 L solution. If the density of CHCl_3 was 1.486 g/mL, the molar concentration of this solution equals:

- A) 3.73×10^{-3} B) 6.21×10^{-3} C) 7.46×10^{-3} D) 5.60×10^{-3}
 E) 8.70×10^{-3}

3. Consider the equation: $\text{Ba}(\text{OH})_{2(aq)} + \text{H}_2\text{SO}_{4(aq)} \rightarrow \text{BaSO}_{4(s)} + 2\text{H}_2\text{O}_{(l)}$, if 380.0 mL of 0.273 M $\text{Ba}(\text{OH})_{2(aq)}$ solution were mixed with 500.0 mL of 0.520 M $\text{H}_2\text{SO}_{4(aq)}$, 20.0 g $\text{BaSO}_{4(s)}$ (233.4 g/mol) were obtained, the % yield of reaction equals:
- A) 67.07 % B) 62.93 % C) 83.60 % D) 75.33 E) 87.73 %

A) B, Ge, Te, Po B) Yb, Sm, Eu, Dy C) Pa, Np, Am, No
D) P, S, Cl, Br E) Sc, Zr, Cd, Os

A) 4, 2, -2, $\frac{1}{2}$ B) 4, 2, 0, $\frac{1}{2}$ C) 4, 1, 0, $\frac{1}{2}$ D) 4, 1, 1, $\frac{1}{2}$ E) 4, 2, -1,

										IIA										IIIA					IVA	VA	VIA	VIIA																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
										4										3					6	7	8	9																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
										Be										B					C	N	O	F																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
										9.0122										10.81					12.01	14.0064	15.9994	18.9984																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
										12										13					14	15	16	17																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
										Mg										Al					Si	P	S	Cl																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
										24.305										26.9815					28.086	30.9738	32.06	35.453																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
										IIIb	IVb	Vb	VIb	VIIb	VIII	IX	Xb	XIb	XIIb	Ib	IIb	IIIb	IVb	Vb	VIb	VIIb	VIIIb	IXb	XB	XIb	XIIb																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																						
										20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210	211	212	213	214	215	216	217	218	219	220	221	222	223	224	225	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255	256	257	258	259	260	261	262	263	264	265	266	267	268	269	270	271	272	273	274	275	276	277	278	279	280	281	282	283	284	285	286	287	288	289	290	291	292	293	294	295	296	297	298	299	300	301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316	317	318	319	320	321	322	323	324	325	326	327	328	329	330	331	332	333	334	335	336	337	338	339	340	341	342	343	344	345	346	347	348	349	350	351	352	353	354	355	356	357	358	359	360	361	362	363	364	365	366	367	368	369	370	371	372	373	374	375	376	377	378	379	380	381	382	383	384	385	386	387	388	389	390	391	392	393	394	395	396	397	398	399	400	401	402	403	404	405	406	407	408	409	410	411	412	413	414	415	416	417	418	419	420	421	422	423	424	425	426	427	428	429	430	431	432	433	434	435	436	437	438	439	440	441	442	443	444	445	446	447	448	449	450	451	452	453	454	455	456	457	458	459	460	461	462	463	464	465	466	467	468	469	470	471	472	473	474	475	476	477	478	479	480	481	482	483	484	485	486	487	488	489	490	491	492	493	494	495	496	497	498	499	500	501	502	503	504	505	506	507	508	509	510	511	512	513	514	515	516	517	518	519	520	521	522	523	524	525	526	527	528	529	530	531	532	533	534	535	536	537	538	539	540	541	542	543	544	545	546	547	548	549	550	551	552	553	554	555	556	557	558	559	560	561	562	563	564	565	566	567	568	569	570	571	572	573	574	575	576	577	578	579	580	581	582	583	584	585	586	587	588	589	590	591	592	593	594	595	596	597	598	599	600	601	602	603	604	605	606	607	608	609	610	611	612	613	614	615	616	617	618	619	620	621	622	623	624	625	626	627	628	629	630	631	632	633	634	635	636	637	638	639	640	641	642	643	644	645	646	647	648	649	650	651	652	653	654	655	656	657	658	659	660	661	662	663	664	665	666	667	668	669	670	671	672	673	674	675	676	677	678	679	680	681	682	683	684	685	686	687	688	689	690	691	692	693	694	695	696	697	698	699	700	701	702	703	704	705	706	707	708	709	710	711	712	713	714	715	716	717	718	719	720	721	722	723	724	725	726	727	728	729	730	731	732	733	734	735	736	737	738	739	740	741	742	743	744	745	746	747	748	749	750	751	752	753	754	755	756	757	758	759	760	761	762	763	764	765	766	767	768	769	770	771	772	773	774	775	776	777	778	779	780	781	782	783	784	785	786	787	788	789	790	791	792	793	794	795	796	797	798	799	800	801	802	803	804	805	806	807	808	809	810	811	812	813	814	815	816	817	818	819	820	821	822	823	824	825	826	827	828	829	830	831	832	833	834	835	836	837	838	839	840	841	842	843	844	845	846	847	848	849	850	851	852	853	854	855	856	857	858	859	860	861	862	863	864	865	866	867	868	869	870	871	872	873	874	875	876	877	878	879	880	881	882	883	884	885	886	887	888	889	890	891	892	893	894	895	896	897	898	899	900	901	902	903	904	905	906	907	908	909	910	911	912	913	914	915	916	917	918	919	920	921	922	923	924	925	926	927	928	929	930	931	932	933	934	935	936	937	938	939	940	941	942	943	944	945	946	947	948	949	950	951	952	953	954	955	956	957	958	959	960	961	962	963	964	965	966	967	968	969	970	971	972	973	974	975	976	977	978	979	980	981	982	983	984	985	986	987	988	989	990	991	992	993	994	995	996	997	998	999	1000	1001	1002	1003	1004	1005	1006	1007	1008	1009	1010	1011	1012	1013	1014	1015	1016	1017	1018	1019	1020	1021	1022	1023	1024	1025	1026	1027	1028	1029	1030	1031	1032	1033	1034	1035	1036	1037	1038	1039	1040	1041	1042	1043	1044	1045	1046	1047	1048	1049	1050	1051	1052	1053	1054	1055	1056	1057	1058	1059	1060	1061	1062	1063	1064	1065	1066	1067	1068	1069	1070	